



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SV FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

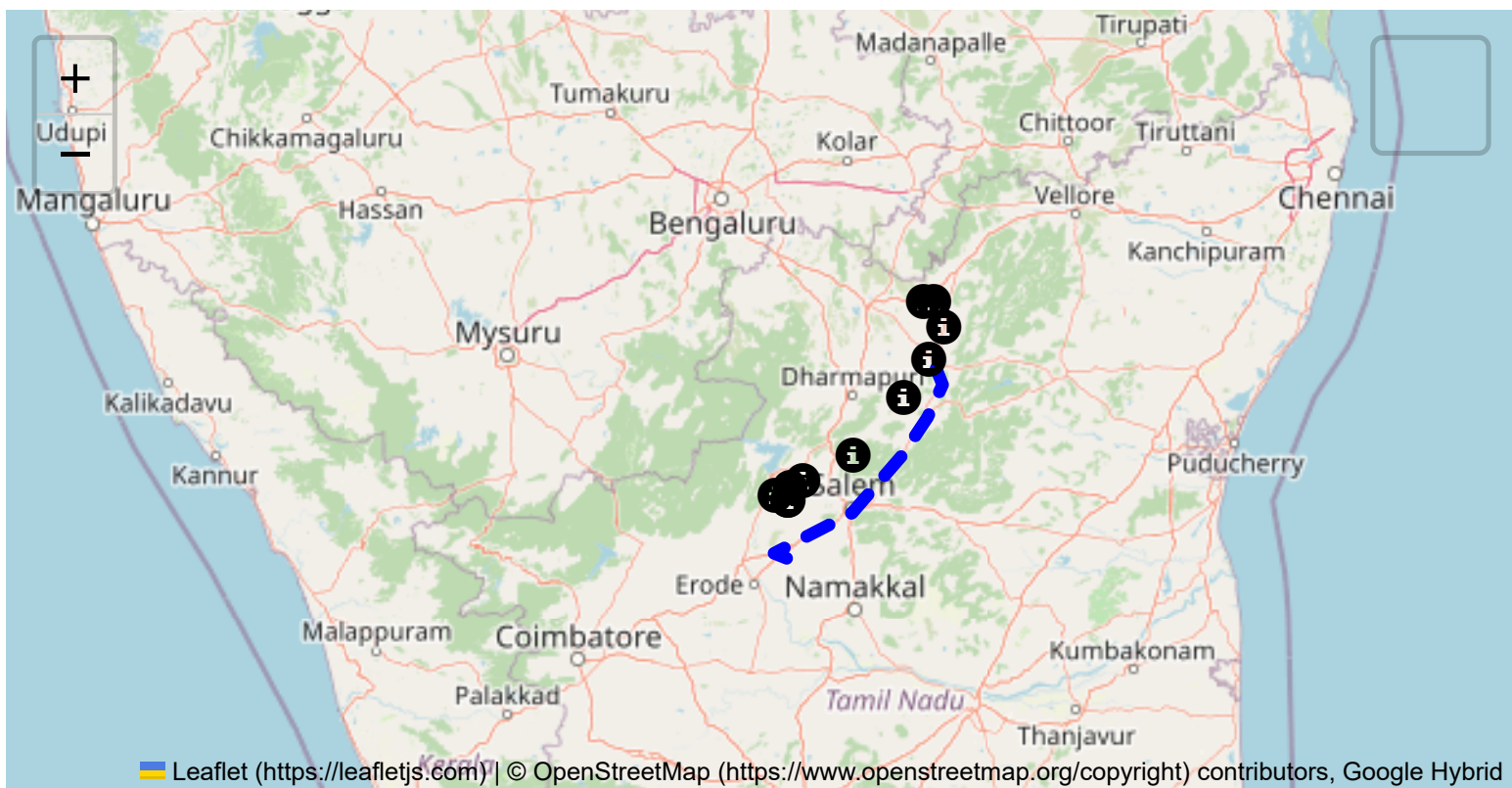
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

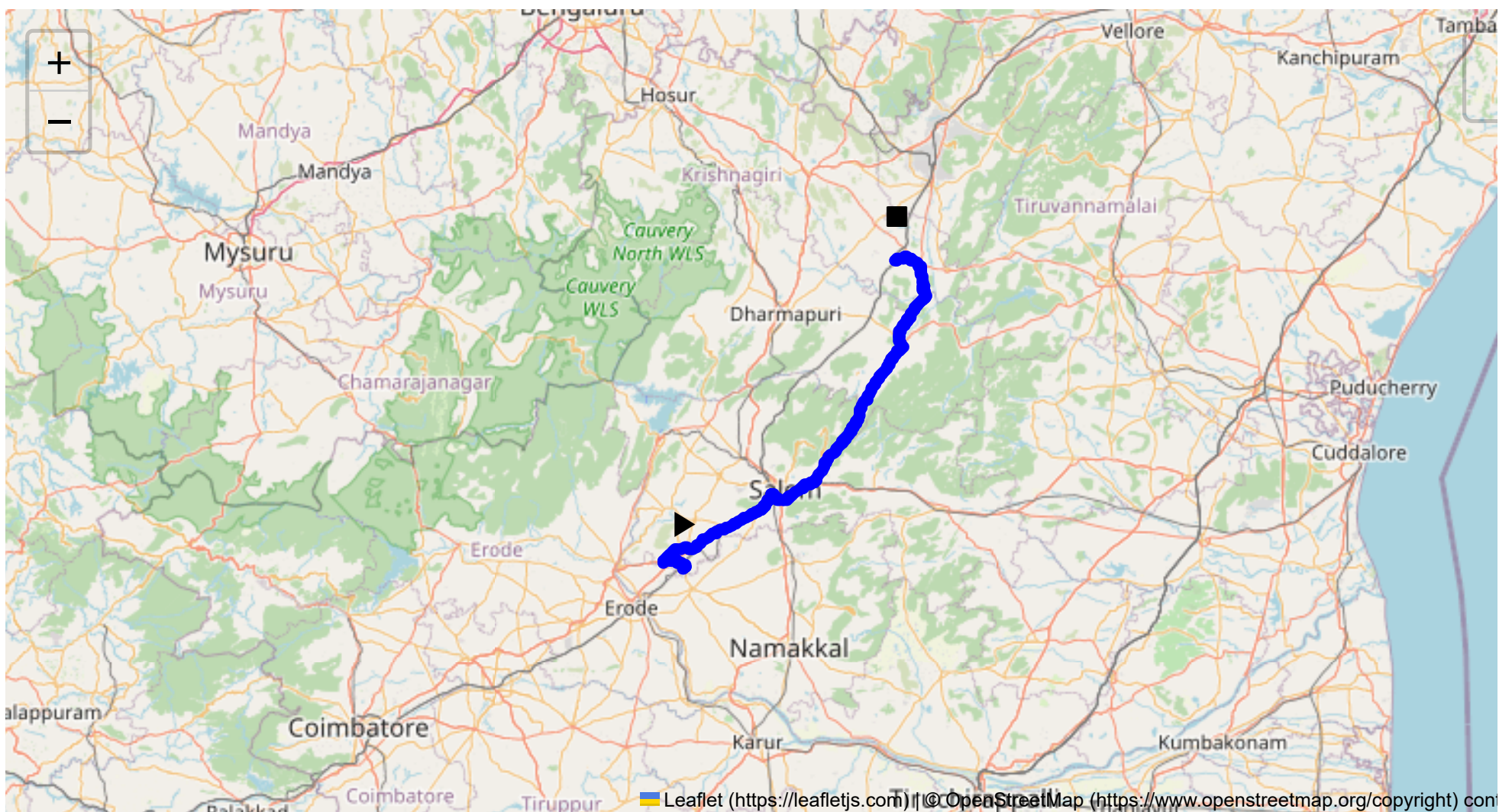
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 153.50 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.3 hours
Start: (11.4381, 77.8734)
End: (12.2816898, 78.4622079)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route spans approximately 153.50 kilometers, beginning in Sangagiri, Tamil Nadu, and ending in Salamarathupatti, Tamil Nadu. It covers both rural and semi-urban areas, with various road types, including local roads, main highways, and bypasses. Key points include Sangagiri, Puthur, Padaiveedu, Salem, Ittalapatti, and Selakarampatti. The route traverses various terrains, including agricultural and semi-industrial landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

The region typically experiences a tropical climate with pronounced wet and dry seasons. Expect higher rainfall during the monsoon months (June to September), which can lead to waterlogging and slippery roads. During the dry season, dust and heat could affect visibility and vehicle performance. Fog can be a hazard in early mornings during the winter months (December to February).

3. Analysis of Traffic Patterns:

Traffic congestion is commonly encountered near urban centers like Salem, particularly during peak hours (8:00-10:00 AM and 5:00-7:00 PM). Sangagiri and other smaller towns may also experience congestion during local market hours. The Salem Bypass is designed to alleviate congestion but can get busy when incidents occur on alternative routes.

4. Assessment of Road Quality and Infrastructure:

The predominant highways, such as Salem-Kochi Hwy and NH 179A, are generally well-maintained, but potholes and roadworks can occasionally hinder the journey. Some local roads in rural sections might be narrower with variable surface quality. Regular maintenance issues and abrupt changes in road conditions should be anticipated.

5. Suggestions for Alternative Routes for Emergencies:

In case of disruptions, alternate routes include taking the Salem-Coimbatore Highway (if heading back towards major hubs) and smaller regional roads that connect to bypasses around Salem. Use real-time traffic data apps to make informed decisions on the go.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

Tamil Nadu enforces strict regulations for transporting hazardous materials. Ensure compliance with at least the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules. Night travel may be restricted, and special permits are usually required. Adherence to signage, route restrictions, and weight limits is crucial.

7. Overview of Historical Incidents:

There have been occasional incidents involving heavy vehicles like rollovers and chemical leaks, especially in Salem where the industrial activity is higher. Investigations revealed that driver fatigue and inadequate vehicle maintenance were common factors.

8. Environmental Considerations and Sensitive Areas:

The route crosses agricultural lands and approaches industrial zones near Salem. Farmers may move livestock and machinery on roads, posing temporary hazards. Protect waterways, as several roads cross rivers and canals. Ensuring no spillage or contamination is vital in these sensitive zones.

9. Analysis of Communication Coverage:

Most of the route has standard cellular coverage, but dead zones may exist in more remote or heavily forested sections between smaller towns. It's advisable to carry emergency communication devices like satellite phones in case of prolonged outages.

10. Estimated Emergency Response Times:

Response times can range from 15–30 minutes in urban areas like Salem to over an hour in remote zones near Ittalapatti, depending on the specific location and time. Emergency services are more readily available near larger towns.

11. Overall Summary of Risk Assessment:

The route presents a moderate level of risk to heavy vehicle operators transporting hazardous materials. Key challenges include navigating urban congestion, variable road conditions, and potential weather-related issues. Adherence to regulations, awareness of local hazards such as dense traffic and rough terrain, and proactive communication strategies will mitigate risks effectively. Continuous monitoring of local news and traffic conditions is essential for maintaining a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Roundabout	High	12.26577, 78.53533	15 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44024, 77.87528	15 KM/Hr
4	Turn	High	11.44898, 77.87410	15 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45764, 77.85219	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45545, 77.81812	30 KM/Hr
9	Turn	High	11.45584, 77.81695	15 KM/Hr
10	Turn	Medium	11.45656, 77.81709	30 KM/Hr
11	Turn	Medium	11.53552, 77.95763	30 KM/Hr
12	Turn	Medium	11.61280, 78.10411	30 KM/Hr
13	Turn	Medium	11.64096, 78.11795	30 KM/Hr
14	Turn	High	11.64210, 78.11860	15 KM/Hr
15	Turn	High	11.64172, 78.12154	15 KM/Hr
16	Turn	Medium	11.64098, 78.12163	30 KM/Hr
17	Turn	High	11.63972, 78.12083	15 KM/Hr
18	Turn	Medium	11.63668, 78.12232	30 KM/Hr
19	Turn	Medium	11.62826, 78.13337	30 KM/Hr
20	Turn	High	11.66372, 78.20649	15 KM/Hr
21	Turn	Medium	11.67124, 78.22961	30 KM/Hr
22	Turn	High	11.67111, 78.23134	15 KM/Hr
23	Turn	High	11.67187, 78.23142	15 KM/Hr
24	Turn	Medium	11.67427, 78.23892	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.81208, 78.34071	30 KM/Hr
26	Turn	Medium	11.83518, 78.35305	30 KM/Hr
27	Turn	Medium	11.84832, 78.36115	30 KM/Hr
28	Turn	Medium	12.05026, 78.48298	30 KM/Hr
29	Turn	Medium	12.05407, 78.48313	30 KM/Hr
30	Turn	Medium	12.07570, 78.47610	30 KM/Hr
31	Turn	Medium	12.17751, 78.54783	30 KM/Hr
32	Turn	Medium	12.17821, 78.54845	30 KM/Hr
33	Turn	High	12.18592, 78.54966	15 KM/Hr
34	Turn	Medium	12.20108, 78.54385	30 KM/Hr
35	Turn	Medium	12.20933, 78.54354	30 KM/Hr
36	Turn	Medium	12.21071, 78.54412	30 KM/Hr
37	Turn	Medium	12.24997, 78.53501	30 KM/Hr
38	Turn	Medium	12.25781, 78.53395	30 KM/Hr
39	Turn	Medium	12.25897, 78.53314	30 KM/Hr
40	Turn	Medium	12.26049, 78.53306	30 KM/Hr
41	Turn	High	12.26736, 78.53058	15 KM/Hr
42	Turn	Medium	12.27375, 78.52657	30 KM/Hr
43	Turn	Medium	12.28246, 78.50937	30 KM/Hr
44	Turn	Medium	12.29036, 78.49815	30 KM/Hr
45	Turn	Medium	12.28953, 78.49238	30 KM/Hr
46	Turn	Medium	12.28824, 78.49140	30 KM/Hr
47	Turn	Medium	12.28516, 78.48467	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
17	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
20	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
24	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
25	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
26	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
33	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
34	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium
35	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium
36	hospital	Government Hospital, Uthangarai	12.26612, 78.535949	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
23	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
27	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
28	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
29	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
30	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
31	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
32	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium
37	school	Dheeran Chinnamalai International Residential School	12.2833281, 78.5072053	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.26577, 78.53533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53552, 77.95763



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61280, 78.10411



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64096, 78.11795



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64210, 78.11860



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.12154



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64098, 78.12163



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63972, 78.12083



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63668, 78.12232



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66372, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67124, 78.22961



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67111, 78.23134



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67187, 78.23142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67427, 78.23892



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.84832, 78.36115



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05026, 78.48298



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05407, 78.48313



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07570, 78.47610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17751, 78.54783



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17821, 78.54845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.18592, 78.54966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20108, 78.54385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20933, 78.54354



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21071, 78.54412



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24997, 78.53501



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25781, 78.53395



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25897, 78.53314



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26049, 78.53306



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.26736, 78.53058

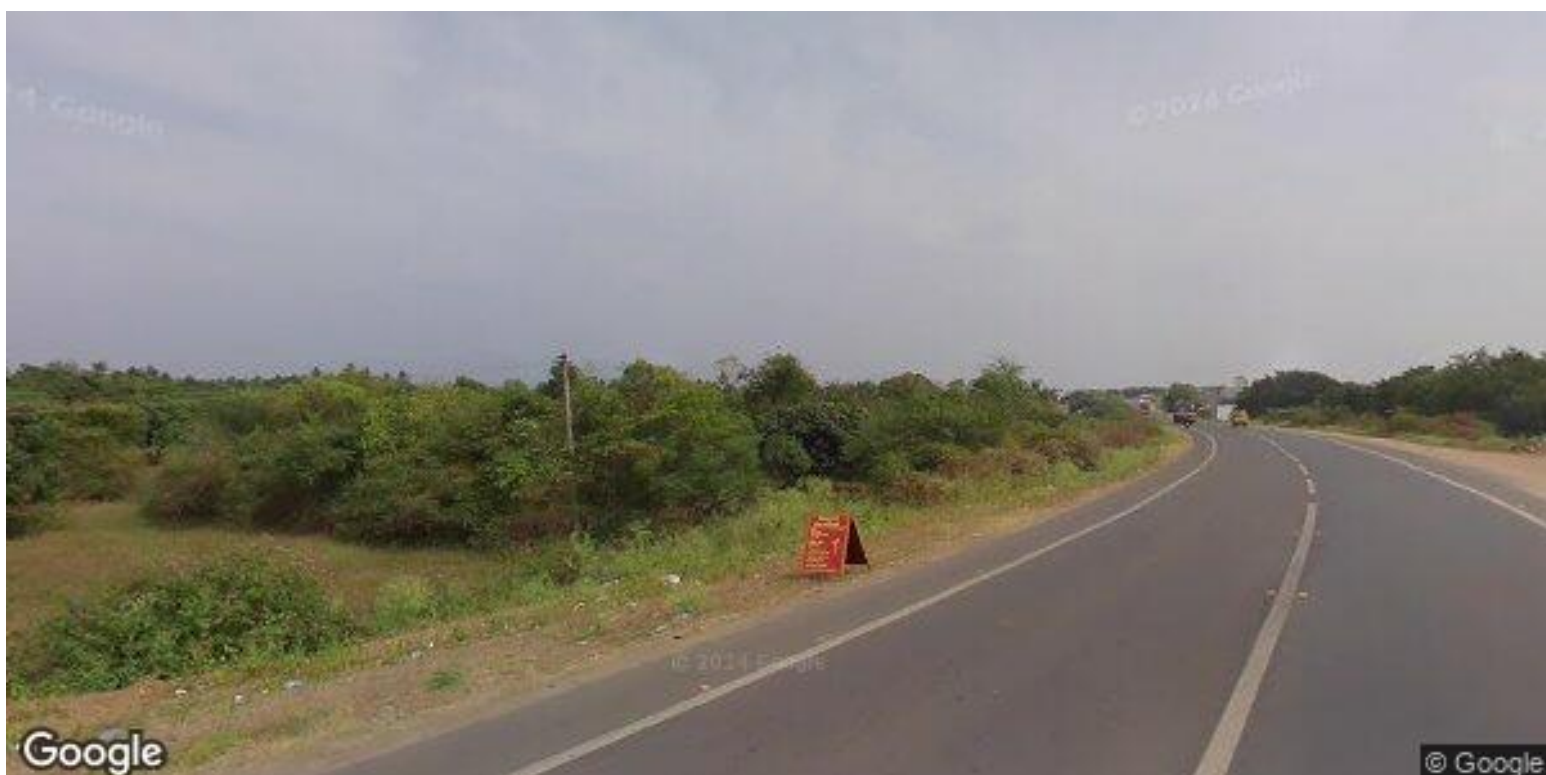


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.27375, 78.52657



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.28246, 78.50937
